



GOA UNIVERSITY

F.E. - (Civil / Computer / Information Technology / VLSI Design & Technology / Mechanical / Electrical / Electronics & Telecommunication) (SEM I & II)(AY-2024-25)

Paper Title: Creative Thinking & Innovation

Paper Code: AEC-151

[Duration: 2 Hours]

[Total Marks: 30]

- Instructions:
- i) Answer any two questions from Part A
 - ii) Answer any two questions from Part B
 - iii) Answer any one question from Part C
- Assume relevant data wherever necessary

Part A

- | | | |
|----|---|----|
| Q1 | a) Karthik, a first year engineering student wants to purchase a Laptop. Help him to make the decision by constructing a checklist for identifying different features that he should take into consideration. | 02 |
| | b) Justify the encouragement provided by corporates for divergent thinking over convergent thinking to come up with innovative solutions for complex problems. | 02 |
| | c) Explain the significance of accurate problem definition. | 02 |
| Q2 | a) Classify the types of innovation studied by you. | 02 |
| | b) Explain the technique of record-discuss-clarify-verify | 02 |
| | c) Contrast the role of technology push with market pull to the success of a product. | 02 |
| Q3 | a) Explain Viceral, Behavioral and Reflective design with an example | 03 |
| | b) Compare any three decision making methods that you have studied. | 03 |

Part B

- Q4 a) Provide an example of a product that effectively utilizes the multifunction principle **02**
b) Give an example of a preliminary action taken in a project to mitigate risks. **02**
c) What are the key sections in the detailed design of a patent? **02**
- Q5 a) Evaluate the role of compensation in balancing trade-offs during product design. **02**
b) Illustrate how recognizing a blessing in disguise can lead to innovative solutions. **02**
c) Explain about the patent acts / patent laws that you need to be aware of when filing a patent. **02**
- Q6 a) Provide an example of segmentation used effectively in product design or manufacturing. **02**
b) Explain how changing symmetry can affect functionality or aesthetics in design. **02**
c) How do you evaluate the potential for patenting an idea? **02**

Part C

- Q7 a) Explain the interaction matrix method of decision making **02**
b) Explain the following: virtual simulation and physical simulation. **02**
c) Explain the technique of force field analysis. **02**
- Q8 a) Illustrate an example where recycling has been effectively integrated into product development. **02**
b) Illustrate an example where opaque and porous materials were used effectively together in design or engineering. **02**
c) Assuming that your innovative idea can be patented and the product will be successfully commercialized, how will you assess that the product has an optimum Useful Life? **02**